

Factors affecting energy requirements and its determinants

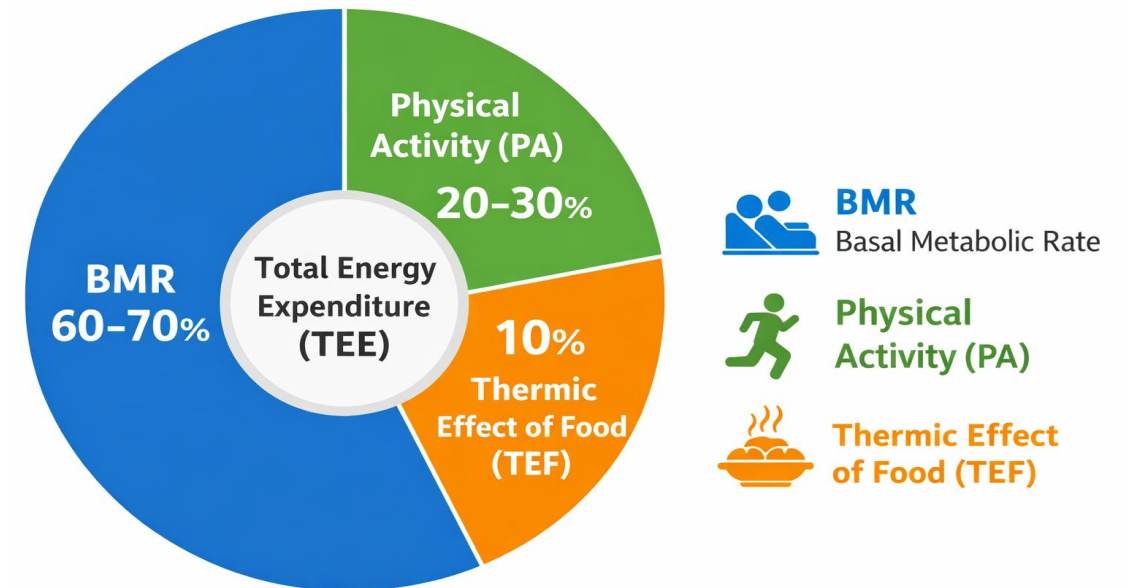
What Is Energy?

- Energy is the capacity to do work
- In nutrition, energy is required for:
 - Basal metabolic functions
 - Physical activity
 - Growth, repair, and maintenance
- Unit of measurement:
 - Kilocalorie (kcal)
 - Kilojoule (kJ)

- **Energy Requirement:** The amount of dietary energy needed to maintain:
 - Body size and composition
 - Normal physical activity
 - Optimal health
 - Expressed as:
 - kcal/day or kJ/day
- Energy requirements are determined by total energy expenditure (TEE), influenced mainly by **Basal Metabolic Rate (BMR)** (body size, composition, age, sex), **Physical Activity (PA)** (intensity, duration, occupation), and the **Thermic Effect of Food (TEF)**, with added needs for growth, pregnancy, lactation, or illness; factors like genetics, health status, and environment also play roles in calculating daily caloric needs for weight maintenance or change.

• Components of Energy Requirement

- Total Energy Expenditure (TEE) consists of:
 - Basal Metabolic Rate (BMR)
 - Physical Activity Energy Expenditure (PAEE)
 - Thermic Effect of Food (TEF)



- **Basal Metabolic Rate (BMR):** REE (Resting Energy Expenditure) - measure the calories your body burns at rest for vital functions.
- The energy needed to maintain basic life functions (breathing, circulation, cell production) in a completely resting state, after a long fast (12-18 hrs) and upon waking, often 60-70% of TEE.
- Energy for basic life functions (breathing, circulation, cell function) at rest,.
- Energy required to maintain vital functions at rest Includes:
 - Breathing
 - Circulation
 - Cell metabolism
- Measured under standard conditions:
 - Fasting state
 - Physical and mental rest
 - Neutral temperature

Factors Influencing BMR

- Age
- Sex
- Body size and composition
- Hormonal status
- Genetic factors

Physical Activity (PA)

- 20–30% of total energy
- Depends on intensity & duration
- Most variable factor

- **Energy spent on:**
 - Occupational work
 - Daily activities
 - Exercise and sports

- **Depends on:**
 - Intensity of activity
 - Duration
 - Frequency
 - Type of occupation (sedentary, moderate, heavy work)

Thermic Effect of Food (TEF)

- **~10% of total energy expenditure**
- Energy required for:
 - Digestion
 - Absorption
 - Metabolism of nutrients
- **Macronutrient effect:**
 - Protein → highest TEF
 - Carbohydrate → moderate
 - Fat → lowest

- **Factors Affecting Energy Requirements**

- 1. Age**

- **Infants & children:** High energy (growth)
- **Adults:** Stable
- **Elderly:** Reduced (loss of muscle mass, lower BMR)

- 2. Gender**

- **Males:** Higher energy needs (greater muscle mass)
- **Females:** Lower BMR, but ↑ needs during pregnancy & lactation

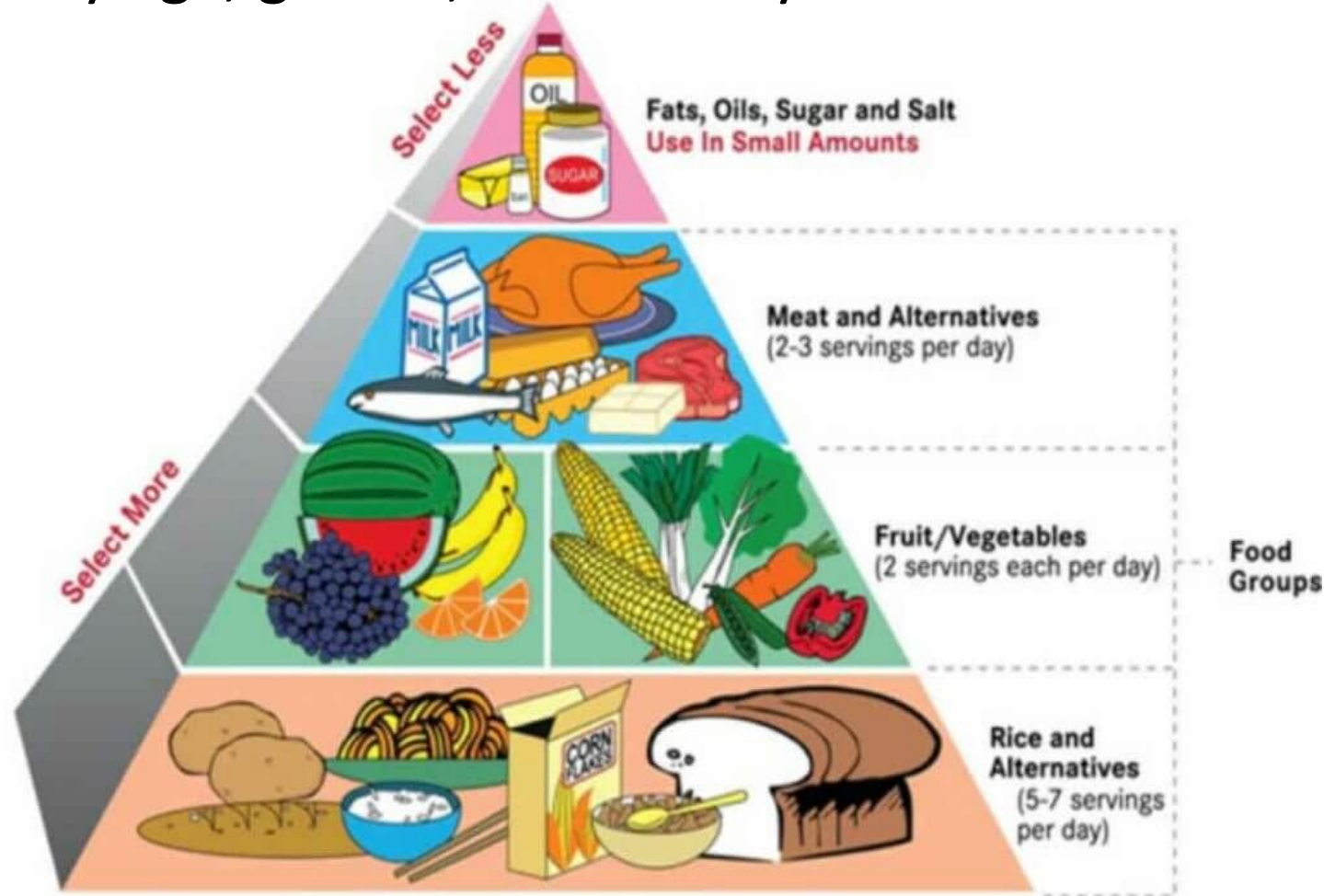
3. Body Size & Composition

- Larger body → higher energy requirement
- More **lean mass** → higher BMR
- Higher fat mass → lower metabolic activity

4. Growth & Physiological State

- Childhood & adolescence
- Pregnancy
- Lactation
- 🙌 Increased energy required for tissue synthesis and milk production

- A balanced daily diet should generally consist of 50–60% carbohydrates, 10–15% protein, and 20–30% fats. A healthy, sustainable daily intake often centers around 2000 calories, though this varies by age, gender, and activity level.



Components of a Balanced Diet Chart

1. Vitamins and Minerals –

- These are body protecting nutrients.
- They regulate body metabolism and protect against infections, deficiency diseases and any metabolic imbalance.
- To include multi-colored, whole fruits and vegetables that are a treasure of minerals and vitamins.



2. Protein

- This nutrient is associated with bodybuilding, growth, and development. This is required to help your body repair cells and make new ones. **whole grains, dals, chickpeas, peas, lentils, beans, peanuts, milk, eggs, fish, and meat.**



3. Carbohydrate

- This nutrient is equally necessary as associated with energy giving and sustainability. The voluntary and involuntary functions of the body are well-regulated with the efficient supply of energy in the form of glucose. Consume complex carbohydrates which include whole grains such as **brown rice, wheat, ragi, oatmeal and potatoes**



Rice

Bread

Chapatis

Potatoes

Bananas

4. Fats

- Fats play an important role in providing energy, circulation of vitamins and hormone synthesis. Include more of fats
- sunflower oil, fish oil, nuts and seeds.

Fats



Butter



Oil



Nuts



Avacados



Meats & Fishes



Chocolates



- **Diet-Induced Thermogenesis (DIT)**

- **Diet-Induced Thermogenesis (DIT)**, also known as the thermic effect of food, is the energy your body uses to digest, absorb, transport, metabolize, and store nutrients after a meal, contributing 5-15% to daily energy expenditure and causing a temporary rise in body heat

- **Key Aspects of DIT**

- **Energy Expenditure:** An increase in metabolism above basal levels after eating, primarily for nutrient processing.
- **Components:** Includes energy for digestion, absorption, transport, and storage, with storage being a significant factor, especially for fats.
- **Macronutrient Influence:** Protein and alcohol generate more DIT than carbohydrates, while fats generate the least.
- **Factors Affecting DIT:** Meal size, macronutrient mix, physical activity, age, sleep, and even chewing speed can alter DIT levels.

- **Factors that influence DIT:**

- **Eat Protein & Alcohol:** Higher protein and alcohol content in meals increases DIT.
- **Chew Thoroughly:** Slow, mindful eating, known as "Fletcherism," increases DIT and can aid weight control.
- **Regular Exercise:** Physical activity may enhance DIT, though findings vary.